



**ALPHA CHOICE INNOVATIVE
ACADEMY, AREPO, OGUN STATE**

THIRD TERM

GRADE 8

BIOLOGY

NAME: _____

SYNOPSIS

WEEK

SKILL

1&2.	FAMILY HEALTH (DISEASES).
3.	REPRODUCTIVE HEALTH.
4.	ABSTINENCE-I
5&6.	ABSTINENCE-II
7&8.	DRUG ABUSE.
9&10.	VARIATION AND CLASSIFICATION.
11.	REVISION
12.	EXAMINATION

WEEK 1&2: FAMILY HEALTH (DISEASES)

OBJECTIVES:

At the end of this lesson, students should be able to:

1. Identify the two major types of diseases.
2. Highlight two vectors and the diseases they carry.
3. Analyze the consequences of contracting diseases or infections.
4. Describe ways of disease control and prevention; and
5. What are disease vectors?

DISEASE: Can be defined as a deviation of the living organism from the normal state of health, which shows the visible symptoms. i.e., an illness, a sickness, an ailment/ is the disruption of normal body functions, which may alter a person's normal work and social activities. Disease occurs when the microorganism, which causes the disease, enters the body and overcomes the body defense, causing illness or unhealthy condition of the body.

TYPES OF DISEASES

When we suffer from one sickness or another, it is said that we have a disease. For example, frequent stooling is usually associated with diarrhoea disease. Generally, if an organism does not have any disease, it is considered to be healthy.

Disease can be grouped into two major classes, communicable diseases and non-communicable diseases.

COMMUNICABLE (INFECTIOUS) DISEASES: are diseases that can be passed from one organism to another. Such diseases are caused by bacteria, viruses etc. Examples of communicable are measles, chicken pox, tuberculosis (TB), etc. These diseases are usually caused by organisms called pathogens.

NON-COMMUNICABLE (NON INFECTIOUS) DISEASES: are diseases which cannot be passed from one organism to another. These diseases are linked with nutrition (e.g. food deficiency) such as kwashiorkor or linked to ageing or environment. Some diseases may also be inherited. Others are cancer, heart disease, ulcer etc.

DISEASE VECTORS

Vectors are agents that pass pathogens from one living organism to another. The vector itself is not affected by the disease. Common examples of disease vectors are insects (mosquito, tsetse fly, house fly, black fly etc.), dogs (rabies), and rats (food poisoning Salmonella)

DISEASE PREVENTION

Some of the ways diseases can be controlled and prevented include proper sanitation, proper health education, knowledge of the life history of the disease-causing micro-organisms, immunization, choice of life partners to control diseases such as sickle cell anaemia and treatment of infected persons.

SANITATION

It is well known that disease-causing organisms thrive in dirty environments. Disease can be prevented by maintaining a clean environment. This can be done in several ways. First of all, it is necessary to drain all stagnant water to prevent mosquitoes and insects from breeding in them. Oil can also be sprayed on stagnant water. Cleanliness of the home- clothes, utensils, floor, toilets, use of disinfectants, etc. all helps to control and prevent the spread of diseases. Clearing and burning of bushes also help to maintain a clean environment and control the spread of diseases.

HEALTH-EDUCATION

Some people lack the awareness of the causes of diseases, how they spread and how they can be prevented. Educating people about these will create the awareness and enable such person to take preventive measures against contracting diseases.

Education plays a major role in the control of diseases. The knowledge of the life history of mosquitoes is a good example of how education can help us stop spread of disease vectors by eliminating the conditions necessary for its growth and development.

IMMUNIZATION

Immunization is the introduction of vaccines into the body to help in fighting disease-causing organisms that may enter the body. Such diseases include measles, poliomyelitis, tetanus, yellow fever, whooping cough and tuberculosis etc. Therefore, the essence of immunization is to stimulate the body to produce antibodies, which give immunity to the body against some diseases

CHOICE OF LIFE PARTNERS (Pre- marital medical screening)

Choice of life partners is another way to prevent or control the risk of diseases. The major example is the case of sickle cell anaemia. Knowledge of the genotype of different persons helps to prevent persons with genotypes that can result in giving birth to sicklers. Also, in the case of indiscriminate sexual relationships creates high risk of being infected with sexually transmitted diseases including HIV/AIDS. The choice of a life partner should be finally considered, based on the health status of each. Prospective husbands and wives are advised to carry out diseases screening tests such as HIV/AIDS, before agreeing to marry each other, Screening will help to eliminate hereditary and sexual transmitted diseases.

TREATMENT

Treatment of an already infected person will help to stop further spread of the disease.

CONSEQUENCES OF DISEASE CONTRACTION

Contraction of disease results in the person being in a state of ill-health or being described as sick. This can have severe consequences on the individual, family and society.

When an individual is sick, the person is not able to carry out his normal activities as a student, worker or others. It also requires the spending of money to buy drugs to take care of the sick person. This is closely related to the family who feels and bears the impact of the sick individual in terms of cost and so on. In some cases, the cost of treatment is so much that it cannot be borne by the family which leads to distress and pain in the family.

All these have an impact on the society at large. There is loss of production, wages and income. Also revenue is spent to provide health and medical services. These negative impacts can be reduced by taking actions.

SUMMARY

- Disease: Can be defined as a deviation of the living organism from the normal state of health, which shows the visible symptoms. i.e. an illness, a sickness, an ailment.
- Types of disease are communicable and non-communicable disease.
- Disease occurs when the microorganism, which causes the disease, enters the body and overcomes the body defence, causing illness or unhealthy condition of the body.

COMMON EXAMPLES OF DISEASE VECTORS ARE INSECTS

Disease	vector
1. Malaria	Female anopheles mosquito-(plasmodium)
2. Sleeping sickness	Tsetse fly-(Trypanosome)
3. Cholera	House fly(Vibro cholerae)
4. River blindness	Black fly-(onchorcerca)
5. Elephantiasis or filariasis	Culex mosquito
6. Yellow fever	Aedes mosquito(virus)
7. Rabies	Dog
8. Food poisoning, lassa fever	Rats(Salmonella)

WAYS OF PREVENTING DISEASES ARE

1. Sanitation,
2. Health Education,
3. Immunization,
4. Choice Of Life Partner And
5. Treatment Of Infected Persons.

CONSEQUENCES OF DISEASE/INFECTION

- -The individual spends money, which could have served other needs, on medicines.
- -The individual becomes less sufficient and less productive at work, home or school.
- -The individual's contribution to the economy of the family and the society becomes less.
- -The family spends more money on medicines, thereby affecting the family budget.
- -The individual spends money, which could have served other needs, on medicines
- -The overall productivity of the society is reduced.
- -The government will spend more on health of the nation.
- -The workforce of the society will be reduced.

STUDY QUESTIONS

Explain the following terms

1. Diseases

2. Diseases vectors
3. Communicable diseases
4. Non – communicable diseases
5. Sanitation
6. Immunization

Explain the transmission process of the following:

7. Malaria
8. Sleeping sickness
9. River blindness

10. Enumerate three measures of controlling the following vectors:

- (a) Mosquito
- (b) Housefly.

WEEK 3: REPRODUCTIVE HEALTH

OBJECTIVES:

At the end of this lesson, students should be able to:

1. Identify the myths about breastfeeding
2. Highlight breastfeeding and its advantages.
3. Outline causes of infant mortality.
4. Explain the meaning of genetic counselling.
5. Define the word reproductive health.

REPRODUCTIVE HEALTH

Is a state of complete physical, mental, social wellbeing and not merely the absence of disease or infirmity in all matters relating to reproductive system.

Reproductive health is the ability to have a satisfying and safe sex life, capability to reproduce and freedom to decide, if, when and how often to do so. Men and women of all ages must care for their reproductive health in order to attain an excellent physical, social and mental wellbeing.

GENETIC COUNSELLING

It is important for parents to find out about the genes in individual family before getting married. Find out if there characteristic disorders in the families. Genetic counselling should be done in a good hospital, where the pregnant will be told about sex determination of a baby and the effects of genetics on a baby's eye, hair, skin, colour, height etc. This will help people with family genetic disorder make decision about having children.

BREAST FEEDING

It is always good to breastfeed babies. Within the first hours after delivery, the mammary gland (breast) begins to secrete milk which is very good for baby.

The breast milk contains nutritional elements and antibodies which give the baby temporary immunity to infectious disease.

The baby is usually breast fed, but when breast milk is not adequate for the baby's needs, the baby may be given some other milk forms in addition. Commonly used

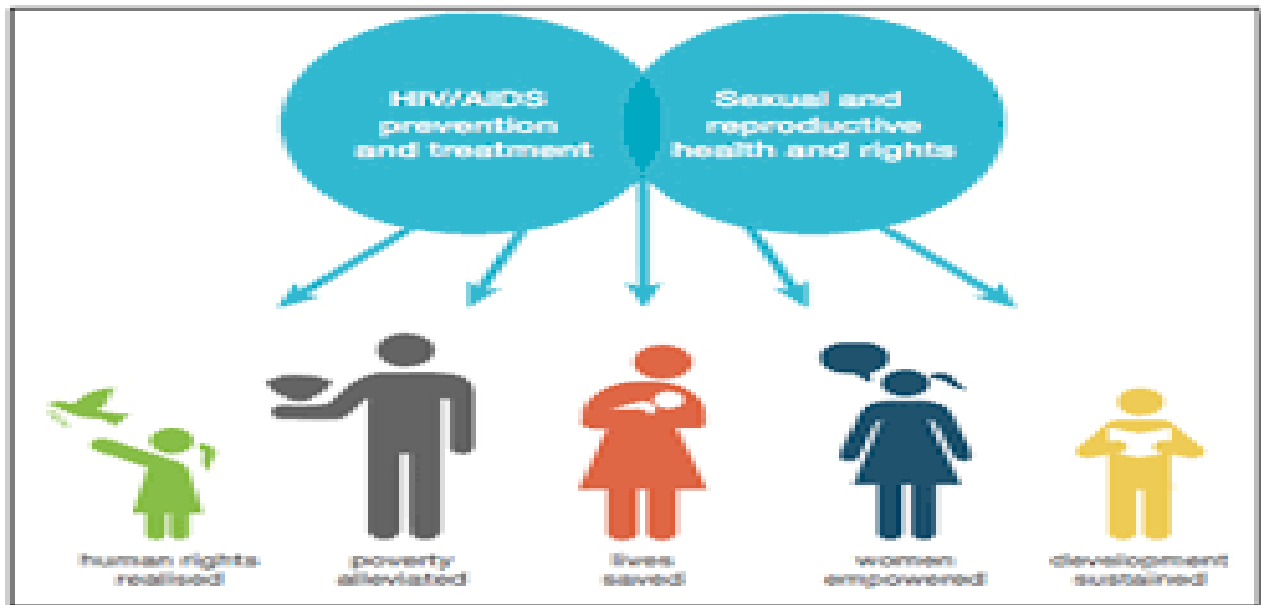
artificial milk is cow's milk prepared in a special way for use by human babies. However, there is no better milk for human babies than human milk in the sense that:

1. It is readily available.
2. It is at normal body temperature.
3. There is no risk of contamination.
4. It helps the uterus/womb back to its normal place soon after delivery.
5. It promotes emotional connection between mother and child.
6. Above all it is very cheap. During breast feeding the nursing mother must be cautious of using alcohol and doctor's unprescribed drugs.

MYTHS ABOUT BREAST-FEEDING

The following statements must be dispelled:

1. Once you are breast-feeding, no pregnancy can occur if you have sexual intercourse. It has been found that some women ovulate during breast feeding, so such a woman can get pregnant.
2. You should not have sexual intercourse when you are breast-feeding so that the baby does not suck sperm in the breast milk. (False)
3. Processed milk is better than breast milk. (False)
4. Breast feeding babies need extra vitamin D.
5. A breast-feeding mother should wash her nipples each time before breast-feeding the baby.
6. It is easier to bottle feed than to breast feed.
7. Breast feeding ties the mother down.
8. If the mother has an infection, she should stop breast feeding.
9. If the baby has diarrhoea/vomiting, the mother should stop breast feeding.
10. If the mother is taking medicine, she should not breast feed.



COUNSELLING AND TESTING FOR HIV

People living with HIV and AIDS require information, counselling, care and support. In general, they should be advised to:

1. Protect them against any form of sexually transmitted infection.
2. Avoid passing the infection to others through unprotected sexual intercourse.
3. Share information about their HIV status with their spouses and those they trust.
4. Eat nourishing food and take plenty of water.
5. Seek counselling about pregnancy for the health of both the mother and the child.

STUDY QUESTIONS

1. What is reproductive health.
2. Mention (5) myths about breastfeeding
3. Highlight (4) importance of breastfeeding
4. What do you understand by infant mortality.
5. Outline (3) causes of infant mortality.
6. Explain the meaning of genetic counselling.
7. Outline three importance of prenatal care.
8. What are the risk factors for birth defects?
9. What is the purpose of counselling people living with HIV
10. Mention (2) preventive measures for HIV.

WEEK 4: ABSTINENCE- (I)

OBJECTIVES:

At the end of this lesson, students should be able to:

1. Identify all the benefits of abstinence.
2. Analyze different types of abstinence.
3. Highlight the disadvantages of abstinence.
4. Explain the meaning of Abstinence

ABSTINENCE: Is also called CONTINENCE; is the practice of refraining oneself from indulging in something, typically alcohol or sex/ is a choice to refrain from sexual activity or drug addict.

BENEFITS OF ABTINENCE

- Is a great way to avoid the risks that come with sex like pregnancy and sexual transmitted diseases, until you are ready to prevent and/ or handle them?
- It is a safe way to comply with religious and moral codes that prohibit/ discourage sexual activity before marriage.
- It can also help you focus on things that are important to you, like friends, school, sports activities, having fun, and planning for your future.

TYPES OF ABSTINENCE

Food, tobacco smoking, alcohol, pleasure, sexual abstinence, religions, organizations.

1. FAST, FASTING: Abstaining from food.
2. DIET, DIETING: The act of restricting your food intake or your intake in particular foods.
3. HUNGER STRIKE: A voluntary fast undertaken as a means of protest.
4. SOBRIETY/ TEMPERANCE: Abstaining from excess.
5. TOETOTALLING (TOETOTALISM): Abstaining from alcohol.
6. SELF CONTROL, SELF DENIAL, AND SELF DISCIPLINE: The act of denying yourself, controlling your impulses.

7. SAWM: The third pillar of Islam is fasting primarily during the month of Ramadan. Muslims abstain from food and drink and gambling and all pleasures from sunrise to sunset during Ramadan.
8. CELIBACY: Is generally recognized as a voluntary choice to remain unmarried/ engage in any form of sexual activity, usually in order to fulfill a religious vow.
9. ABSTENTION: Abstaining from sexual relations as because of religious vows.
10. INHIBITION, SUPPRESSION (PSYCHOLOGY): The conscious exclusion of unacceptable thoughts or desires.

DISADVANTAGES OF ABSTINENCE

1. Many abstainers feel like they are the only people not having sex (although this is far from the truth).
2. People may find it difficult to abstain for long periods of time. Women and men often end their abstinence without being prepared to protect themselves against pregnancy or infection.

STUDY QUESTIONS

1. What are the skills and behaviours that promote abstinence?
2. What is the difference between celibacy and abstinence?

What is the meaning of the following?

3. Family planning.
4. Birth defects.
5. Sexual transmitted diseases
6. Dieting
7. Hunger strike
8. Temperance
9. Toe totalism
10. Self-discipline

WEEK 5&6: ABSTINENCE-(II)

OBJECTIVES:

At the end of this lesson, students should be able to:

1. Identify dangers in engaging in premarital sex and the benefits from abstaining from premarital sex.
2. Highlight reasons why young people must abstain from premarital sex.
3. Explain reasons why young people engage in premarital sex.
4. Define the term abstinence.

What is abstinence?

- The conscious decision not to participate in sexual activity of any kind



REASONS WHY YOUNG PEOPLE ENGAGE IN PRE-MARITAL SEX

- **PRESSURE:** From parents, friends, peer group, lecturer, and future partners i.e. for instance, some male bosses in places of work do mount pressure on their female workers, they want girls that can work for them and still satisfy them sexually.
- **CURIOSITY:** They thought they were searching for reality, but they ended up destroying themselves. They are not satisfy with what their parents, pastors and Christian friends told them concerning sex, they want to experience in themselves.

- **ELECTRONIC MEDIA:** Television, film, radio and video has contributed to the high rate of pre-marital sex. What youths watch on screen determine their behaviour and character.
- **BOOKS AND MAGAZINES:** Some satanic authors are in town destroying the youths, they write some sexual stories, books and magazines, and they bring out many pictures that stimulate the youth to think about sex always.
- **ENVIRONMENTAL INFLUENCE:** We live in a corrupt society where people do not see anything bad in ungodliness, they do not see premarital sex as sin; they see it as a normal thing. Hence, girls are encouraged to dress exposing their bodies.
- **COVETOUSNESS:** Greed for money, wealth and position is another cause of premarital sex. Some ladies want money at all costs hence they are ready to use their bodies to get it by sleeping around with men.
- **INDISCIPLINE:** Lack of indiscipline has led many singles into the dungeon of premarital sex.
- **WRONG ASSOCIATION:** This has led many youths into destruction. Show me your friends and I will tell you who you are.
- **IGNORANCE:** Lack of good sex education has led many youths into premarital sex. Some went into it not knowing what they were doing.
- **WRONG INFORMATION:** Since parents have refused to educate their children on the subject and the church is not forthcoming with anything meaningful, the youths have resorted to any information they can get from anywhere either wrong or right.
- **BROKEN HOMES:** Children from broken homes can fall into wrong hand due to the sanitation of their homes.
- **BAD PARENTING:** Children that are not well brought up are likely to fall into wrong hands.
- **IDLENESS:** An idle hand is the devils workshop. If you are idle, it may cuddle the devil.
- **LONELINESS:** Some claim they went into fornication due to loneliness.



REASONS WHY YOUNG PEOPLE MUST ABSTAIN FROM PRE-MARITAL SEX.

- It devalues the future intimacy of your marriage.
- It leaves you empty and searching for real love.
- It creates resentment and frustration.
- It sets a person on a path of unfulfilling sexual experiences.
- It prevents the greatest intimacy in marriage.
- It breaks God laws and dishonors him.
- It presents huge emotional risk- a physical risk disease and illnesses are rampant among those who engage in this life style.
- It creates a spiritual/ emotional bond without commitment.
- It destroys trust.
- It creates children without strong homes.
- It attempts to shortcut God's plan for marriage and family.
- It is awkward, guilt ridden, unfulfilling and not representative of God's original intent.
- It is disappointing at the physical, emotional and spiritual levels.
- It enlarges sexual desires and makes them insatiable.
- It feeds the abortion industry.

STUDY QUESTIONS

1. Outline three methods that can protect you against HIV/AIDS infection.
2. Explain three consequences of irresponsible sexual behaviours among youths.
3. Highlight three sexual transmitted diseases/infection that can be contracted by an individual through sexual intercourse.
4. Define the term abstinence.
5. Identify dangers in engaging in premarital sex and the
6. List the benefits from abstaining from premarital sex.
7. Highlight (5) reasons why young people must abstain from premarital sex.
8. Explain reasons why young people engage in premarital sex.
9. Highlight (3) ways you can't get infected with HIV.
10. What is the significance of abstinence.

WEEK 7&8: DRUG ABUSE

OBJECTIVES:

At the end of this lesson, students should be able to:

1. Identify the major causes of drug abuse.
2. Highlight different types of drugs.
3. Describe common ways youths misuse drug.
4. Explain the meaning of Drug Abuse.

WHAT IS A DRUG?

A drug is any substance, apart from food that is taken into our body and, because of its nature affects the structure or functions of the body of the person that takes it.

WHY DO WE TAKE DRUG?

It is to fight against diseases in our body, relieve suffering, and save lives.

DRUG ABUSE: Is the act of excessive intake of drugs for a purpose different from what they are meant for or taken for the right purpose, but in a wrong way.

When a drug is misused, that is, when it is not taken according to the instruction given by the doctor, it is used for a purpose it is not meant for, or used for the appropriate purpose but in a wrong dosage, and then the drug has been abused. Drug abuse is dangerous because every drug is made from chemicals. These chemicals can be harmful to some organs of the body when the correct dosage is not used. Some victims turn into beggars, while some have damaged brains and turned to lunatics. Sometimes drug abuse victims die of the effects of the drugs.



CLASSIFICATION/ TYPES OF DRUGS

ITEM	CLASS	EXAMPLE	MEDICAL USE	SIDE EFFECT WHEN ABUSED
1.	Narcotics	1. Opium 2. Morphine	1. To kill/relieve pain from diseases/accidents. 2. To reduce hunger, thirst, sex drive.	When inhaled or smoked, it causes drowsiness, loss of the senses, and renders one helpless.
2.	Tranquillizer	Valium	1. Modifies the state of the emotional/ mood. 2. Reduces symptoms of mental illness.	Dizziness and excess sleep.
3.	Sedatives	Barbiturates	1. To induce sleep. 2. To treat ulcer, hypertension, convulsion, epilepsy and mental disorder.	Weakness, tremors, convulsion, disorientation, dependent on the drug always
4.	Hallucinogen	Marijuana (cannabis, weed)	No medical use	These are drugs that bring about distorted personality, this include thinking, perception and self-awareness.
5	Stimulants	1. Caffeine. 2. Cocaine.	They make the nervous system work above normal level	Increase the heart beat and blood pressure, overactive, irritable, uncoordinated thinking (depression) and death.

CAUSES OF DRUG ABUSE

The common causes include the following:

1. QUEST FOR CONFIDENCE/LACK OF SELF-CONFIDENCE: Many youths, even adults, are not confident to handle issues like addressing a crowd, approaching the opposite sex, and teaching people of the same class. Such people take drugs to induce confidence in them-selves.

2. **PEER INFLUENCE/BAD FRIENDSHIP:** Attitude of copying mates that take drugs.

3. **QUEST TO OVERCOME OR REDUCE STRESS AND FRUSTRATION:** Examples of stress, or issues that may lead to frustration, are dismissal from work, rape, broken marriage, disappointment, failure in business, and crime. The victims start taking drugs to overcome such problems.

4. **QUEST FOR EXCITEMENT.**Such as sport and leisure in civilizing process

5. **CURIOSITY:** Desire to find out how one will feel after taking drugs.

COMMON WAYS YOUTHS MISUSE DRUGS

DRUG MISUSE: is the taking of a drug for its medically-intended purpose, but not in the appropriate amount, frequency, strength, or manner/If a medically prescribed drug is taken inappropriately. Misuse of drugs may:

1. Taking drugs to induce sleep without seeking for prescription from qualified medical personnel.



2. Taking drugs to gain confidence or boldness to commit crimes such as suicide, rape, assault, hurting someone or oneself, among others.

3. When a patient shares a prescribed drug with a friend or family member for whom the drugs was not prescribed.

4. Taking drugs to attain full sexual satisfaction.

5. When a patient misunderstands the directions for use of prescribed drugs. For example, not taking the drugs as stipulated times, periods, or intervals.

STUDY QUESTIONS

Explain the following terms

1. Drug
2. Drug Abuse
3. Drug Misuse
4. Identify the major causes of drug abuse.
5. Highlight different types of drugs.
6. Describe common ways youths misuse drug.
7. Describe three effects of excessive use of drugs.
8. Mention four health problems associated with drug abuse.
9. Outline two substances you can buy in a shop which can alter the action of the body and a person's behavior.
10. Is there a medical usage of cocaine?

WEEK9&10: VARIATION AND CLASSIFICATION

OBJECTIVES:

At the end of this lesson, student should be able to:

1. Identify different types of variation.
2. Highlight different environmental causes of variation.
3. Describe application of variation.
4. Explain the meaning of the word 'variation'.

The millions of types of living things can be organized into categories, where we group together organisms with similar features. This is called the classification system.

Variation between individuals in a species can be continuous or discontinuous, inherited or environmental.

Species

A **species** is a category within the classification system. Living things of the same type belong to the same species. For example, humans are one species and dogs are another species.



A tiger

Individuals of the same species can reproduce to make more individuals of the same species. Two individuals belonging to different species cannot normally reproduce together. If they do, their offspring is usually infertile and unable to reproduce.

For example, animals called ligers are produced when a male lion and a female tiger reproduce. But a liger cannot have offspring. This means that lions and tigers are different species.

What is variation?

All people are human. They belong to the same species. Your friends and classmates may have different eye colour and hair colour. Some will be boys and some will be girls. Some will be tall and some will be shorter. The presence of differences between living things of the same species is called variation.

Variation between different species is always greater than the variation within a species.

Continuous & discontinuous

Some of the features of the different organisms in a species show continuous variation, and some features show discontinuous variation.

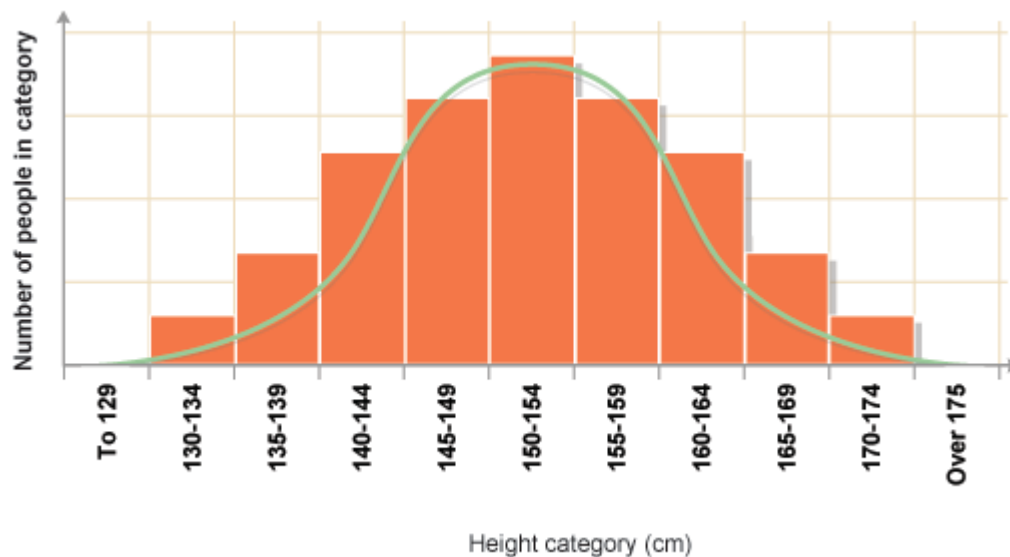
Continuous variation

Human height is an example of continuous variation. Height ranges from that of the shortest person in the world to that of the tallest person. Any height is possible between these values. So it is continuous variation.

For any species a characteristic that changes gradually over a range of values shows continuous variation. Examples of such characteristics are:

- height
- weight
- foot length.

If you record the heights of a group of people and draw a graph of your results, it usually looks something like this:



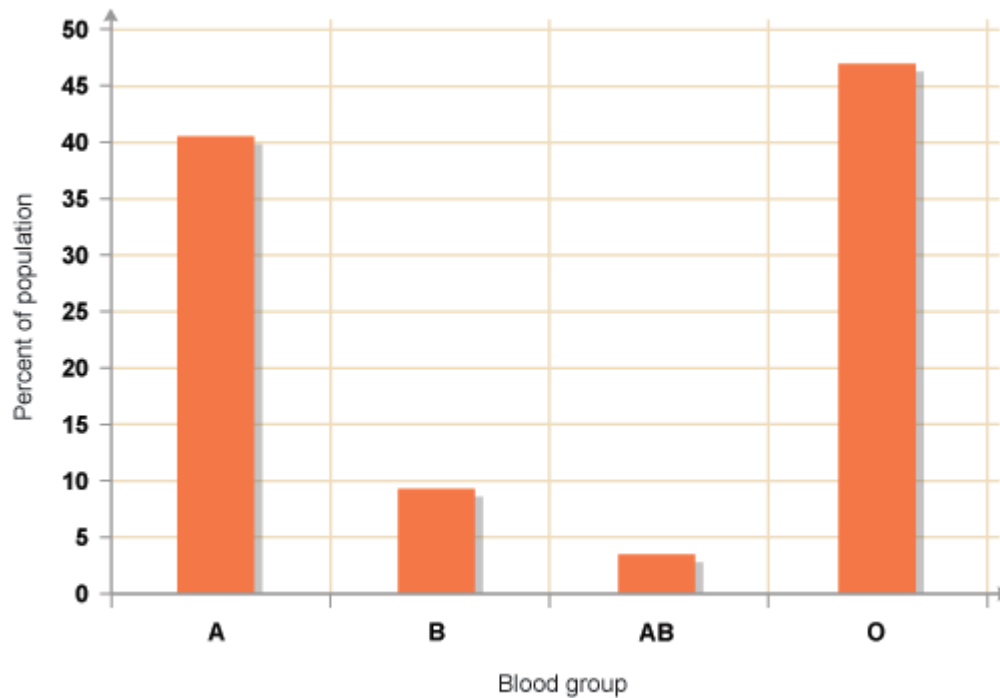
The more people you measure, and the smaller the categories you use, the closer the results will be to the curved line. This shape of graph is typical of a feature with continuous variation. Weight and foot length would give graphs similar in shape to this.

Discontinuous variation

Human blood group is an example of discontinuous variation. There are only 4 types of blood group. There are no other possibilities and there are no values in between. So this is discontinuous variation.

A characteristic of any species with only a limited number of possible values shows discontinuous variation. Here are some examples:

- gender (male or female)
- blood group (A, B, AB or O)
- eye colour.



Inherited & environmental

Some variation within a species is inherited, and some variation is due to the environment.

Inherited causes of variation

Variation in a characteristic that is a result of genetic inheritance from the parents is called **inherited variation**.

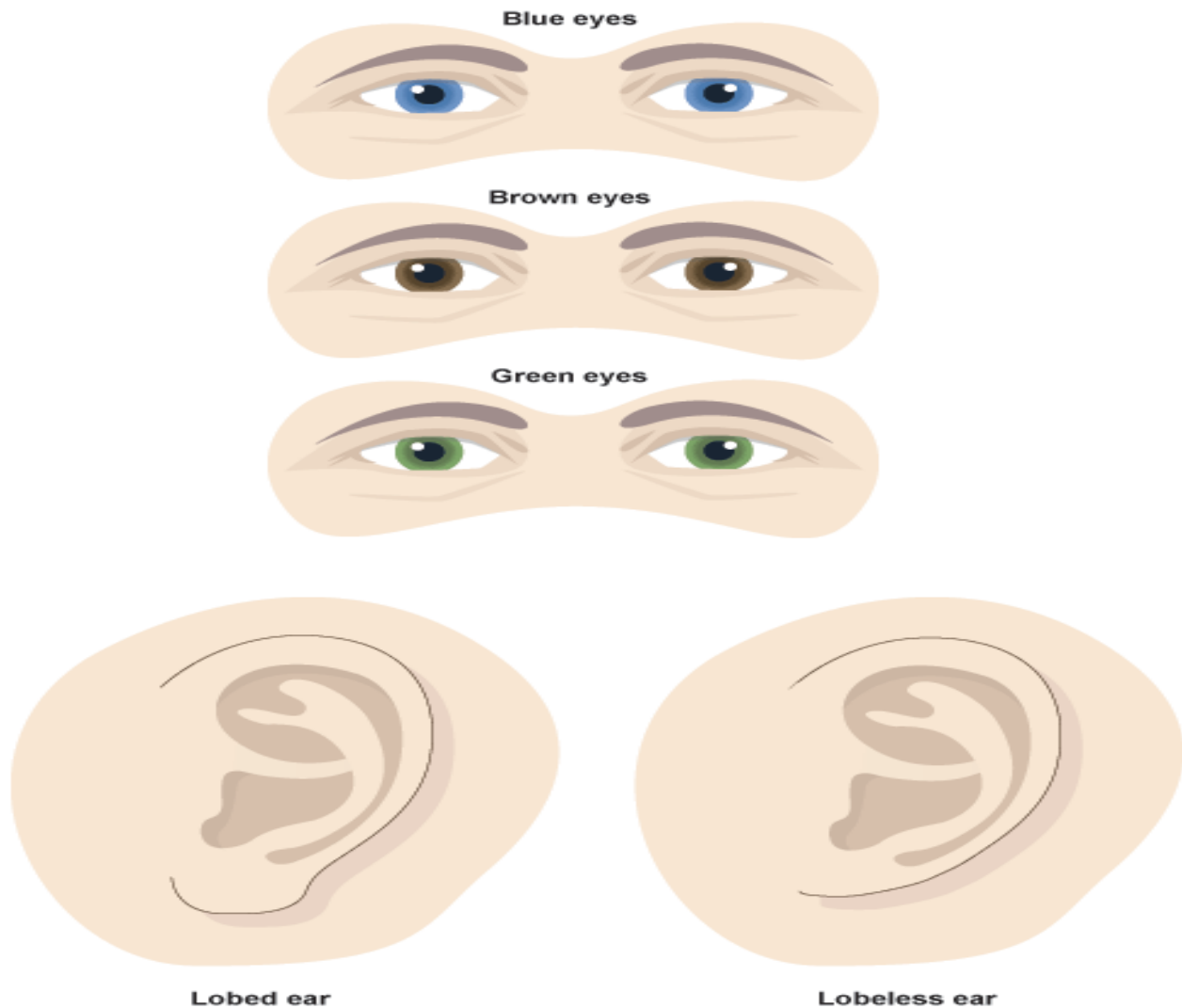
Children usually look a little like their father, and a little like their mother, but they will not be identical to either of their parents. This is because they get half of their inherited features from each parent.

Each egg cell and each sperm cell contain half of the **genetic information** needed for an individual. When these join at fertilization a new cell is formed with all the genetic information needed for an individual.

Here are some examples of inherited variation in humans:

- eye colour
- hair colour
- skin colour
- lobed or lobe less ears.

Gender is inherited variation too, because whether you are male or female is a result of the genes you inherited from your parents.



Environmental causes of variation

Characteristics of animal and plant species can be affected by factors such as climate, diet, accidents, culture and lifestyle. For example, if you eat too much you will become heavier, and if you eat too little you will become lighter. A plant in the shade of a big tree will grow taller as it tries to reach more light.

Variation caused by the surroundings is called **environmental variation**. Here are some other examples of features that show environmental variation:

- your language and religion

- flower colour in hydrangeas - these plants produce blue flowers in acidic soil and pink flowers in alkaline soil.

Both types together

Some features vary because of a mixture of inherited causes and environmental causes. For example, identical twins inherit exactly the same features from their parents. But if you take a pair of twins, and twin 'A' is given more to eat than twin 'B', twin 'A' is likely to end up heavier.

Variety within a species

Natural selection

Within a population of animals, plants or any living organisms, there will be **inherited variations**. Within each species the individuals with the variations best suited to the environment will survive better than the others. More of them will survive to reproduce than the others. When they do, they pass on the genetic information for these variations to their offspring.

Species gradually evolve in this way. This process is called **natural selection**.

Over time a population can change so much it may even become a new species, unable to reproduce successfully with individuals of the original species.

Artificial selection

Artificial selection is when people use **selective breeding** to produce new varieties of a species. A variety is a type of a particular species that is different in some clear way from other varieties of that species.

For example, pedigree dogs come in lots of different varieties, called breeds of dog. They may be different colours and sizes, but they are all still dogs. They are all still the same species. Different varieties of dog have been produced by selective breeding.



Different breeds of dogs

SELECTIVE BREEDING OF COWS

Suppose you wanted a variety of cow that produced a lot of milk. This is what you could do:

- choose or select the cows in your herd that produce the most milk
- let only these cows reproduce
- select the offspring that produce the most milk
- let only these offspring reproduce
- keep repeating the process of selection and breeding until you achieve your goal.

OTHER EXAMPLES OF SELECTIVE BREEDING

The key here is to identify the feature you want, and only breed from the individuals that have that feature. Here are some examples of what selective breeding can produce:

- hens that lay big eggs of a particular colour
- cattle that produce lots of meat
- tomato plants that produce lots of tomatoes
- crops that are resistant to certain plant diseases.

GENETIC ENGINEERING

Variation in living organisms can also be created by **genetic engineering**, also called **genetic modification**, or just 'GM'.

Using laboratory techniques scientists can alter the genetic code within the DNA of a living organism. They can add genes from a different species to an organism's DNA

For example, GM can be used to alter the DNA in a bacterium so that it produces insulin. This is a human hormone and valuable to people with diabetes. Bacteria reproduce quickly, so a lot of insulin can be made quickly and used to help people suffering from diabetes.

Ethical questions

Some people are concerned that there may be negative aspects to GM.

For example, there are people that believe GM is unethical and should be banned.

Other people are concerned that there may be unexpected long-term problems of using GM, producing changes that could not be reversed. Such changes could create new pathogens or "superweeds" by the GM organisms breeding with other species. This could cause the natural balance of living organisms in the environment to be upset.

Scientists, society and individuals need to think about the advantages and disadvantages of genetic engineering.

CLASSIFYING ORGANISMS

There are millions of species on our planet. It would be difficult if we just tried to describe and name each one individually. Although species can be very different from each other, many of them have similar features that allow us to put them into groups.

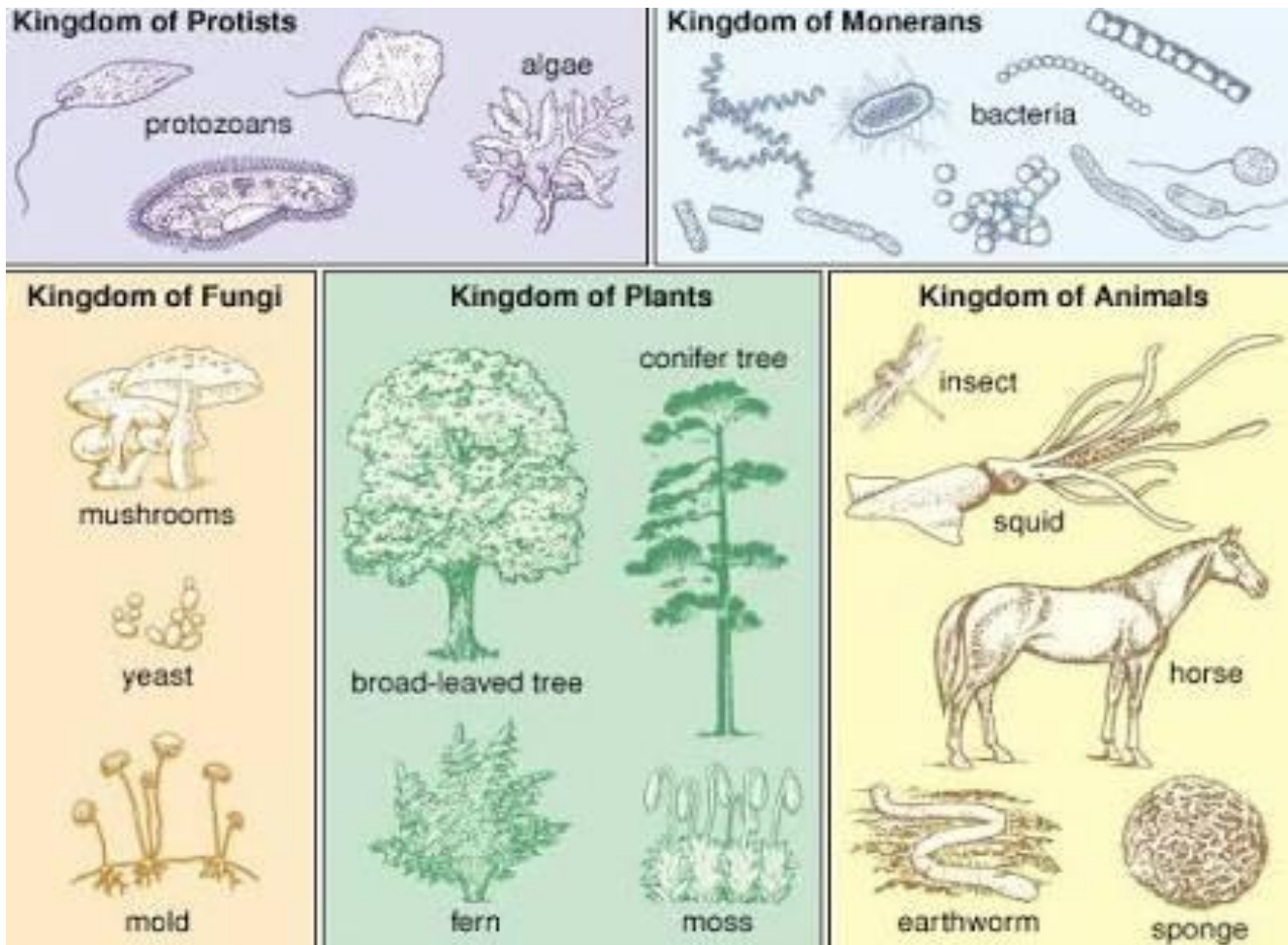
Modern classification system

In the eighteenth-century Carl Linnaeus started the modern system of putting species of organism into certain groups and giving them scientific names. Each species is given a name using Latin words, so that the same name can be used all over the world.

For example, the scientific name for human beings is 'homo sapiens'. Putting different species into different groups according to their features is called **classification**.

The five kingdoms

The first big division of living things in the classification system is to put them into one of five **kingdoms**. These are based on what an organism's cells are like. The table shows the names of the kingdoms and examples of the sort of organisms they contain.



All multicellular animals, including: jellyfish, worms, arthropods, molluscs, echinoderms, fish, amphibia, reptiles, bird (Kingdom Animalia)

STUDY QUESTIONS

1. Boys and girls belong to the same _____.

- ☒ species
- ☐ offspring
- ☐ gender

2. Which of these features is an example of continuous variation?

- ☒ Gender
- ☐ Blood group
- ☐ Weight

3. Which of these features is an example of discontinuous variation?

- ☐ Length of forearm
- ☐ Blood group
- ☐ Height

4. Which of these features shows inherited variation?

- ☐ Brown eyes
- ☐ Having a scar
- ☐ Pierced ears

5. Kathryn has bright blue hair. What kind of variation has caused this?

- ☐ Inherited variation
- ☐ Environmental variation
- ☐ Inherited and environmental variation

6. How much of the genetic information needed for an individual does an egg cell contain?

- ☐ All the genetic information
- ☐ Half of the genetic information
- ☐ None of the genetic information

7. How much of the genetic information needed for an individual does a sperm cell contain?

- ☐ All the genetic information
- ☐ Half of the genetic information
- ☐ None of the genetic information

8. What is one of the processes involved in selective breeding?

- ☐ Choosing individuals from different species to breed together
- ☐ Letting individuals reproduce without interference
- ☐ Choosing individuals with desirable features

9. What does the kingdom 'fungi' include?

- ☐ Mushrooms
- ☐ Algae
- ☐ Ferns

10. What is one difference between conifers and mosses?

- ☒ Conifers are plants but mosses are not
- ☒ Conifers make seeds but mosses do not
- ☒ Conifers don't need water but mosses do not